# Aryan Shah

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## PROFILE

Senior at duPont Manual's MST Magnet with 4+ years of iGEM and wet lab research experience. Earned 2 gold and 1 silver medal on the international stage, along with multiple publications. Outside the lab, I captain a VEX Robotics team, work on independent engineering projects, and spend my free time designing things, breaking things, and figuring out how they work. I'm looking for programs and opportunities that reward my skill set.

## EDUCATION

**duPont Manual High School — MST Magnet** Aug. 2024 – May 2027
*Mathematics, Science & Technology Magnet Program* Louisville, KY

- **GPA:** 3.48 (Unweighted) **ACT:** 32 **SAT:** 1450 **AP Scholar with Honor**
- **AP Courses:** AP Computer Science Principles (3), AP Seminar (3), AP Human Geography (4), AP World History (4), AP Calculus BC (3), AP Physics 1 Algebra Based (3), AP English Language & Composition (3), AP US History (3)
- **Extracurriculars:** VEX Robotics (Captain/President), Technology Student Association (VP), iGEM (Vice President & Promotional Lead), Science Olympiad, League of Legends E-Sports, Science Bowl, Quick Recall, Creative Science Initiative, Ultimate Frisbee

**Jefferson Community & Technical College — Dual Enrollment** Aug. 2024 – May 2025
*Java Programming* Louisville, KY

- Completed college-level Java as a sophomore; applied object-oriented design and algorithm fundamentals directly to Discord bot development and workflow automation projects built shortly after

**Campbellsville University — Dual Enrollment** Aug. 2025 – May 2026
*Spanish 111, 112* Campbellsville, KY

- Completed college-level Spanish coursework as a junior alongside full-time high school studies

**University of Louisville — Dual Enrollment** Aug. 2026 – May 2027
*English 102* Louisville, KY

- Enrolled in college-level composition coursework as a senior, continuing a track record of dual-enrollment coursework alongside high school

## RESEARCH EXPERIENCE

**Louisville iGEM — HS-Kentucky Team** Jan. 2026 – Present
*Vice President & Promotional Lead* Louisville, KY (Hybrid)

- Identifying candidate gene targets for PFAS-binding proteins to engineer an *E. coli*-based whole-cell biosensor capable of detecting PFAS contamination, extending the reverse protein-ligand screening pipeline developed during the 2025 season
- Leading wet-lab execution, experimental design, and team documentation for the 2026 competition season, drawing on three prior years of iGEM experience

**S.Y.A.L.I.S** Nov. 2025 – Aug. 2026
*Scientific Researcher — Intern* Remote

- Earned selection as **1 of 157 from 2,000+ applicants** (sub-8% acceptance) for the 2025 Research Cohort — competing against students from universities and programs worldwide
- Drives independent and collaborative research across multiple disciplines; produces literature reviews, designs experiments, and authors formal research papers under expert mentorship

**Louisville iGEM — HS-KY Team** Jan. – Dec. 2025
*Team Member* | Gold Medal — iGEM Grand Jamboree, Paris Louisville, KY (Hybrid)

- Executed RNA-Seq and Differential Scanning Fluorimetry (DSF) protocols to characterize protein destabilization caused by PFOA — a “forever chemical” linked to cancer and liver disease found in 97% of Americans’ bloodstreams
- Applied reverse-screening across multiple protein-ligand databases to pinpoint candidate human proteins predicted to bind PFOA, forming the backbone of a low-cost biosensor that rivals mass spectrometry and chromatography
- **Co-authored peer-reviewed publication** with the University of Louisville; work presented and awarded Gold at the international Grand Jamboree
- Engineered CAD prototypes, designed Figma web pages, and produced the full suite of promotional graphics deployed across the team’s international competition campaign

**Louisville iGEM — GCM-KY Team** Jan. – Nov. 2024
*Team Member* | Gold Medal — iGEM Grand Jamboree, Paris Louisville, KY (Hybrid)

- Advanced wet-lab skills through a full competition season of hands-on experimentation; built Virtual Cell (V-Cell) computational models to simulate biochemical reaction networks and stress-test experimental hypotheses
- Represented the team in Human Practices meetings, engaging community stakeholders to ground the project in real-world ethical and social context

**Louisville iGEM — Manual-KY Team** Jun. – Nov. 2023
*Team Member* | Silver Medal — iGEM Grand Jamboree, Paris Louisville, KY (Hybrid)

- Competed internationally in the world’s premier synthetic biology competition as a **first-year high school student** with no prior research experience — earning a Silver Medal in the first attempt
- Conducted wet-lab experiments, built V-Cell models, researched project literature, and organized fundraising events to finance team travel to Paris

## PUBLICATIONS

**Predicting and Validating PFOA Binding to Human Proteins** Aug. 2025

- Shah, A. et al. — University of Louisville in partnership with iGEM Louisville
- Identified and validated human proteins that bind PFOA using reverse protein-ligand screening and DSF; demonstrated a purified DNA repair protein is significantly destabilized by PFOA — advancing development of an affordable, accessible alternative to mass spectrometry-based PFAS detection

**Systematic Review of Ubiquilin (UBQLN1-4) Expression in Breast Cancer: Prognostic, Mechanistic, and Therapeutic Impli**

- Shah, A. et al. — Registered with PROSPERO, the International Prospective Register of Systematic Reviews (CRD420251164726)
- Systematic review protocol examining Ubiquilin family (UBQLN1-4) expression patterns in breast cancer and their prognostic, mechanistic, and therapeutic implications

**Transcriptional and Computational Biosensing Pathway Development for the Detection of Per- and Poly-fluoroalkyl Substa**

- Shah, A. et al. — iGEM Louisville, presented as a conference poster at the INBRE Conference, University of Louisville
- Presented computational and transcriptional pathway design work for PFAS biosensing, building on the team’s ongoing PFAS detection research

## CONFERENCES ATTENDED

**KY-INBRE Annual Research Conference** 2024, 2025, 2026

- **2024 (Manual-KY):** Delivered a 10-minute oral presentation on the team’s iGEM project
- **2025 (GCM-KY):** Presented a poster and won **Best Public Health Poster** and **Highest Overall Scoring Poster**
- **2026 (HS-Kentucky):** Presented a poster and won **Best Biomedical Engineering Poster**

**iGEM Grand Jamboree** Paris, France | 2024, 2025

- **2024 (GCM-KY):** Presented the team’s project at the international synthetic biology competition and earned a **Gold Medal**
- **2025 (HS-KY):** Presented the team’s PFOA-binding biosensor research and earned a **Gold Medal**

## CERTIFICATIONS

**Marketing with Canva** — Canva Jul. 2026

- Credential ID: b3fa16

**GE Aerospace Explore Engineering** — Forage May 2026

- Credential ID: 7E9Fd2nBJSmYQoppP

## AWARDS & HONORS

**Gold Medal — iGEM Grand Jamboree, HS-KY Team** Paris, France | Oct. 2025

- Recognized for exceptional wet-lab science, human practices outreach, and project execution at the world's largest synthetic biology competition; third consecutive international medal

**Gold Medal — iGEM Grand Jamboree, GCM-KY Team** Paris, France | Oct. 2024

- Earned second consecutive international medal; recognized for strong experimental methodology, community engagement, and scientific rigor

**Silver Medal — iGEM Grand Jamboree, Manual-KY Team** Paris, France | Oct. 2023

- Medaled internationally as a first-year competitor and first-year high school student — the beginning of three consecutive years on the Paris podium

**Innovate Award — VEX V5 Kentucky State Championship** Mar. 2025

- Awarded for outstanding creativity and ingenuity in robot design and engineering strategy (Team 74656A) at the Kentucky State competition

**Science Fair — Regional 2nd Place & State 3rd Place** 9th Grade, 2024

- Placed 2nd at Regional and 3rd at State Science Fair in the Cellular & Molecular Biology category as a freshman

**Technology Student Association (TSA)**

- **11th Grade:** 3rd Place Forensic Science — State | 2nd Place Forensic Science — Regionals
- **10th Grade:** 3rd Place Manufacturing Prototype — Regionals
- **9th Grade:** 2nd Place Forensic Science — Regionals

**Science Olympiad**

- **9th Grade — State:** 3rd Codebusters, 3rd Detector Building, 5th Flight, 8th Fermi Questions
- **9th Grade — Regionals:** 5th Codebusters, 5th Detector Building **Kentuckiana Invitational:** 4th Flight, 6th Detector Building, 6th Fermi Questions
- **10th Grade — Kentuckiana In-Person Invitational:** 3rd Forensics, 4th Optics, 5th Entomology, 5th Geologic Mapping

**Quick Recall — Undefeated JV Champions** 9th Grade, 2024

- Anchored a JV team that went undefeated through both the JV Challenge and JV League Tournament as a freshman

**AP Scholar with Honor** College Board, 2025

- Recognized for average AP score $\geq$ 3.25 with scores of 3 or higher on four or more AP Exams

## LEADERSHIP EXPERIENCE

**VEX Robotics — duPont Manual High School** 2024 – Present
*Club President & Team Captain* Louisville, KY

- Directs all aspects of team **74656S** — strategy, robot build, and autonomous programming — while managing club operations, member onboarding, and fundraising to keep the program running competitively
- Authored the engineering notebook for team **74656A** through the full 2024–25 season, documenting every design iteration and decision with precision — earning the **2025 Kentucky State Innovate Award**
- Mentors first- and second-year members through the engineering design cycle, building a pipeline of technically capable teammates year over year

**Synthica** Mar. – Aug. 2026
*Discord Developer & Server Moderator* Remote

- Architected and deployed a suite of custom Discord bots, webhook pipelines, and spreadsheet-integrated automation systems that power daily operations for a **1,000+ member global research community**
- Built and maintains custom websites with dynamic content generators for member onboarding, event announcements, and community management
- Enforces community standards across a server spanning dozens of countries, maintaining a professional environment for student researchers at every level

**Synthica Research Journal** Apr. 2026 – Present
*Associate Journal Editor — Biology* Remote

- Evaluates biology manuscript submissions for scientific validity, methodological rigor, and clarity; delivers structured written feedback that guides student authors through the revision process
- Upholds the academic standards of Synthica's publication pipeline, helping student researchers produce work worthy of formal publication

**Technology Student Association — duPont Manual HS** 2024 – Present
*Vice President* Louisville, KY

- Coordinates competition preparation, recruits and orients new members, and runs chapter meetings to keep duPont Manual's TSA chapter competitive at regional and state levels
- Helped grow chapter participation while simultaneously competing and medaling in Forensic Science at both regional and state competitions

## PROFESSIONAL EXPERIENCE

**Synthica** Jan. 2026 – Present
*Growth Department Member* Remote

- Joined as a founding Growth member; engineers and publishes content across LinkedIn to build Synthica's global brand among student researchers
- Scaled the Synthica Discord from **zero to 1,000+ members in under 14 days** — driven by targeted content, community strategy, and consistent outreach
- Leads campaigns for the NSRI Global Research Challenge and Synthica Journal launch, positioning Synthica as the go-to platform for students who want to publish real research without institutional gatekeeping

**KroniQ AI & Solace** Jan. – May 2026
*Growth Marketer / Growth Intern* Remote

- Executed launch content strategies for two early-stage startups back-to-back — Solace (Jan–Mar) and KroniQ AI (Mar–May) — driving user acquisition and brand awareness for each platform during its critical launch window
- Developed platform-specific messaging, produced social content, and built audience engagement from the ground up at both companies

## COMMUNITY SERVICE & VOLUNTEER EXPERIENCE

**Plant Care for Elderly Neighbors** 2022 – Present

- Maintains gardens and waters plants for elderly neighbors in the local community who can no longer do so independently; a service repeated consistently for three-plus years

**Food Packing Volunteer** — Hindu Temple of Louisville 2022 – Present

- Packs and distributes meals at the temple's events, serving families and individuals

**Science Outreach & Promotional Design** — Louisville iGEM 2023 – Present

- Designs outreach materials and organizes public events to introduce synthetic biology to Louisville community members outside of academia — making cutting-edge science legible and exciting for general audiences

**Research Access Advocate** — Synthica 2026 – Present

- Connects students worldwide with free mentorship, project opportunities, and publication pathways — driven by the belief that research experience shouldn't depend on your zip code or your school's name

**Tree Planting Volunteer** — Louisville Grows 2021 – 2024

- Planted trees across Louisville neighborhoods over three years as part of organized urban reforestation efforts to improve green coverage and air quality citywide

**Food Packing Volunteer** — Lifeline Christian Mission 2021 – 2024

- Packed and distributed meals for food-insecure families across the world for three years

## SKILLS

**Laboratory** : RNA Sequencing (RNA-Seq), Differential Scanning Fluorimetry (DSF), protein purification & characterization, V-Cell computational modeling, protein-ligand database screening, gel electrophoresis

**Engineering & Technology** : VEX V5 robot build & programming, CAD / 3D design, Java, Discord API & bot development, webhook & spreadsheet automation, Figma (UI/UX & web design)

**Communication & Marketing** : Scientific writing & peer review, graphic design (print & digital), social media growth strategy, launch campaign execution, public science communication

**Languages** : English (fluent), Hindi (native/bilingual), Spanish (intermediate)